

Name: Zaid

Section: BS-AI-1A

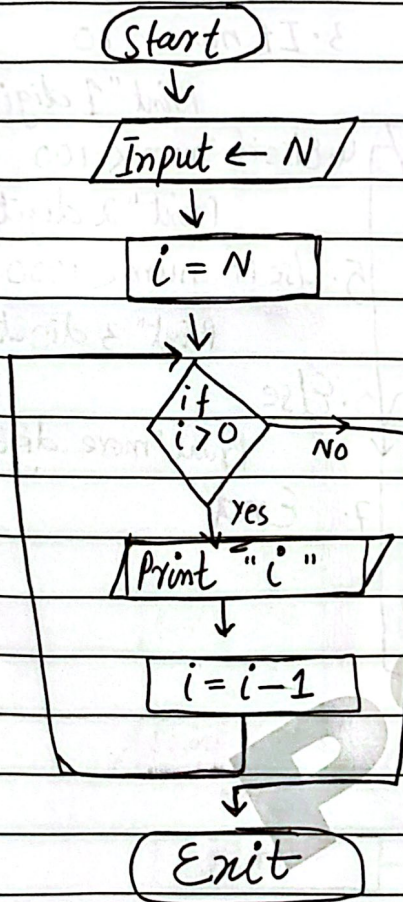
Date: Roll No: 0015

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Foundational Practice Problems

* Question 1

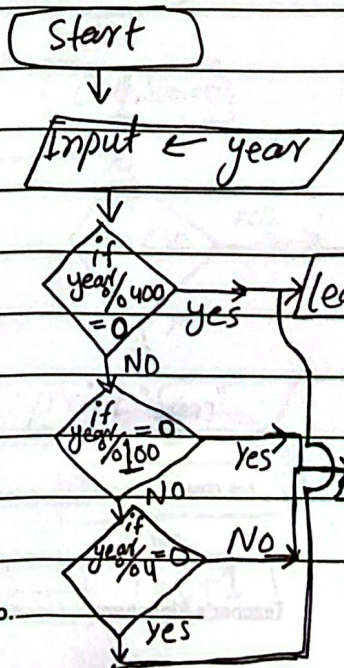
Flow Chart



Pseudocode (Input N)

1. Begin
2. $i = N$ (i is equal to N)
3. while $i > 0$
 Print i
 $i = i - 1$
End while
4. Exit / End

* Question 2



* Pseudocode (year)

1. Begin
2. if $\text{year} \% 400 = 0$
 Print "leap year"
3. Else if $\text{year} \% 100 = 0$
 Print "Not L.Y"
4. Else if $\text{year} \% 4 = 0$
 Print "L.Y"
5. Else
 Print "Not L.Y"
6. End

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Question 3:- (for whole no:s)

★ Pseudocode

(Start)

1. Begin

2. Input (Number = Num)

3. If num < 10

Print "1 digit"

4. Else if num < 100

Print "2 digit"

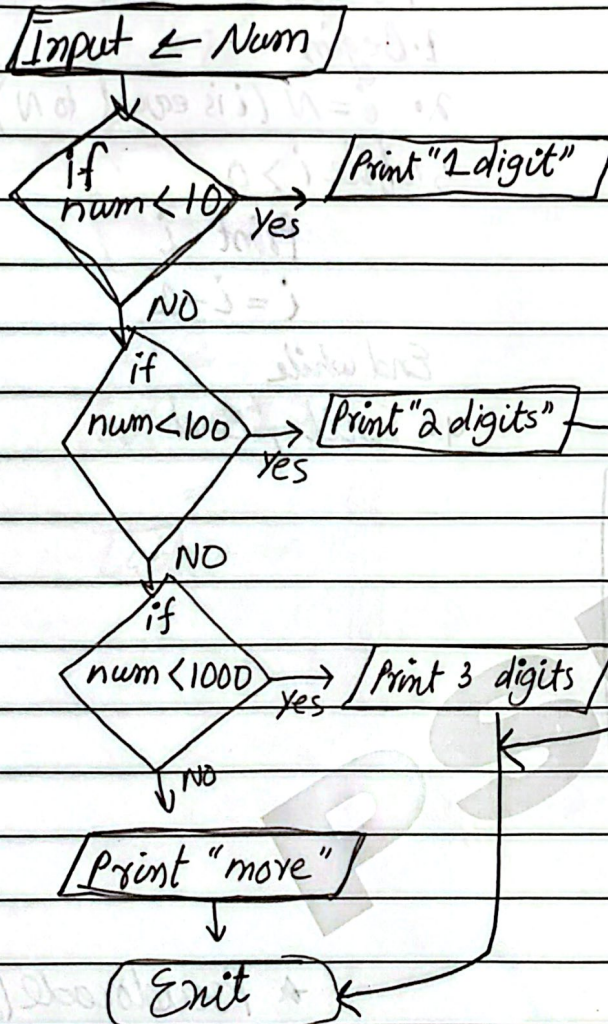
5. Else if num < 1000

Print "3 digit"

6. Else

Print "more digit"

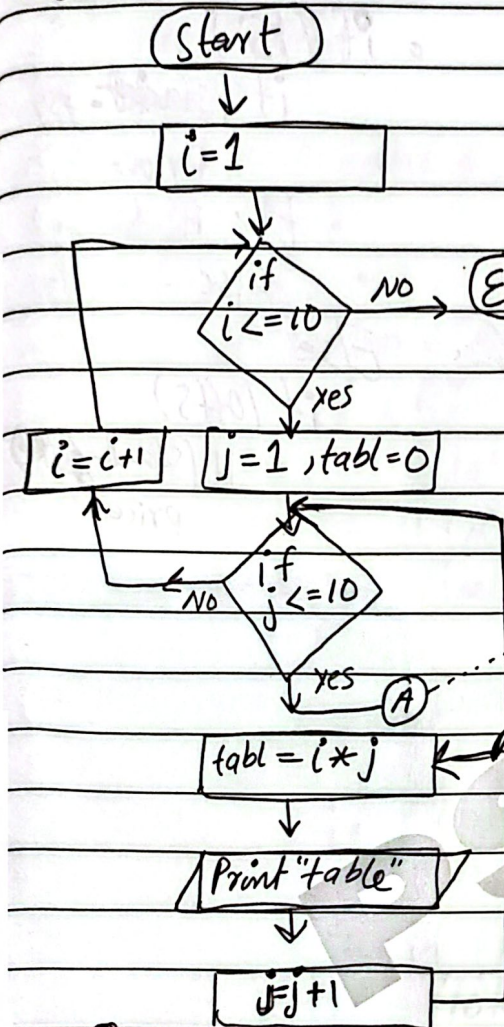
7. End



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Question 4:-



1. Begin

2. $i=1, j=1, tabl=0$

3. while $i \leq 10$;

while $j \leq 10$

if $i \% 3 = 0$

Break/continue

$tabl = i * j$;

Print "table"

$j = j + 1$

$i = i + 1$

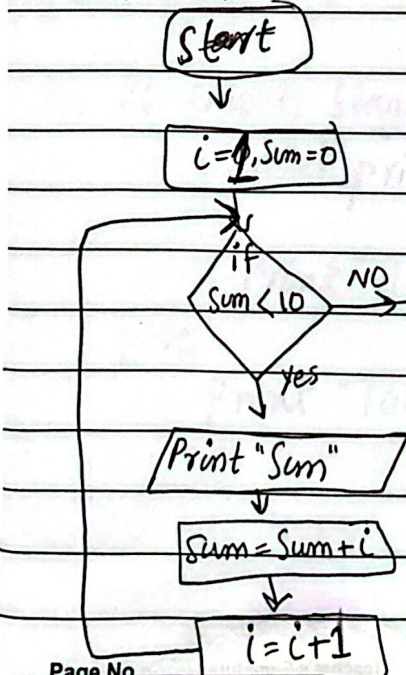
End

P/B Add

if $i \% 3 = 0$

Break

Question 5:



Pseudocode:

1. Begin

2. $i=1, Sum=0$

3. while $Sum < 10$

Print "Sum"

$Sum = Sum + i$

$i = i + 1$

4. End

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* Question 2:

Pseudocode

1. Begin

2. $i = 1, j = 1$

3. While $i \leq N$

Input Num, $\overline{TP}$, S, ~~PS, OS~~, N, RT, j

if (Season = "Peak")

if (RT = "Suite")

$\overline{TP} = 12000 * \text{Nights}$

Else if (RT = "Deluxe")

$\overline{TP} = 8000 * \text{Nights}$

Else

$\overline{TP} = 5000 * \text{Nights}$

Else

if (RT = "Suite")

$\overline{TP} = 8000 * \text{Nights}$

Else if (RT = "Deluxe")

$\overline{TP} = 5000 * \text{Nights}$

Else

$\overline{TP} = 3000 * \text{Nights}$

$j + 1$

while ($j \leq N$)

if (Nights > 7)

$\overline{TP} = \overline{TP} - \overline{TP} * \frac{15}{100}$

Print ($\overline{TP}$)

Else

Print ($\overline{TP}$)

~~4. End~~ ~~Print~~ $j + 1$

4. End

Num
* $\overline{TP}$ = no. of Guests

• $\overline{TP}$ = total price

• S = Season

• PS = Peak Season

• OS = OFF Season

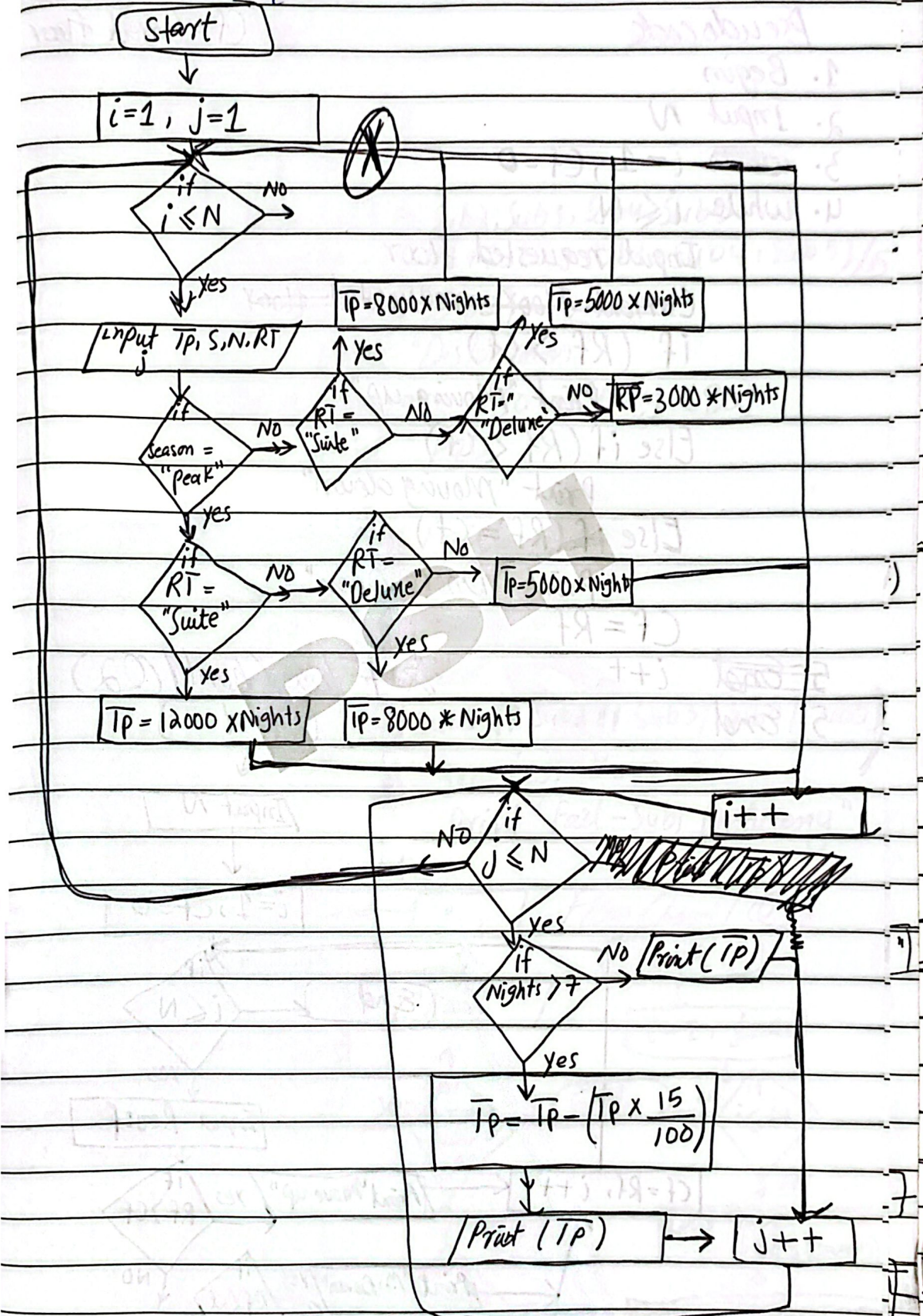
• N = Nights Stayed

• RT = Room type.

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Flow Chart (Question 1)



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* Question #02

Pseudocode

* Rf = Requested floor
 Cf = Current floor

1. Begin
2. Input N
3. ~~while~~ $i = 1, Cf = 0$
4. while $i \leq N$

Input requested floor

~~Current floor = requested floor~~

if $(Rf > Cf)$

Print "Moving up"

Else if $(Rf < Cf)$

Print "Moving down"

Else if $(Rf = Cf)$

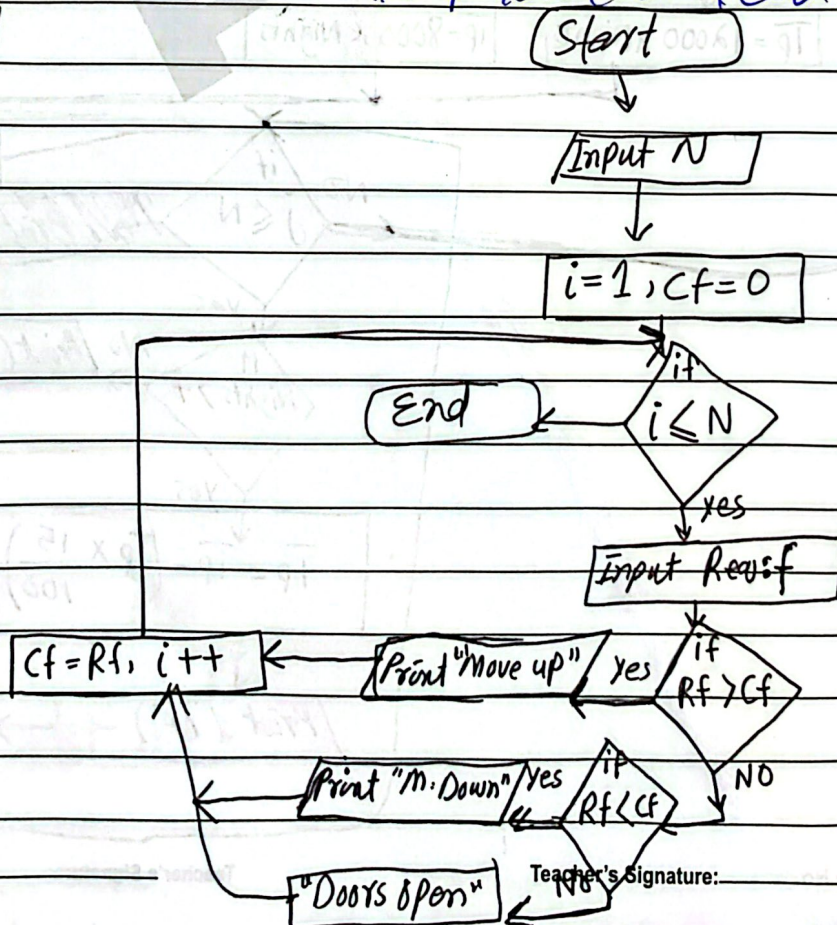
Print "Doors Open"

$Cf = Rf$

~~5. End~~ $i++$

5. End

* Flowchart (Q2)



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Question #03Pseudocode

1. Begin

2. ~~while~~ $i = 1, j = 1$ 3. while $i \leq N$

Input Sub1, Sub2, Sub3, Sub4, Sub5

Average = $(\text{Sub1} + \text{Sub2} + \text{Sub3} + \text{Sub4} + \text{Sub5}) / 5$ if (Average ≥ 80)

Print "Distinction"

Else if (Average ≥ 60 or Avg < 80)

print "Pass"

Else

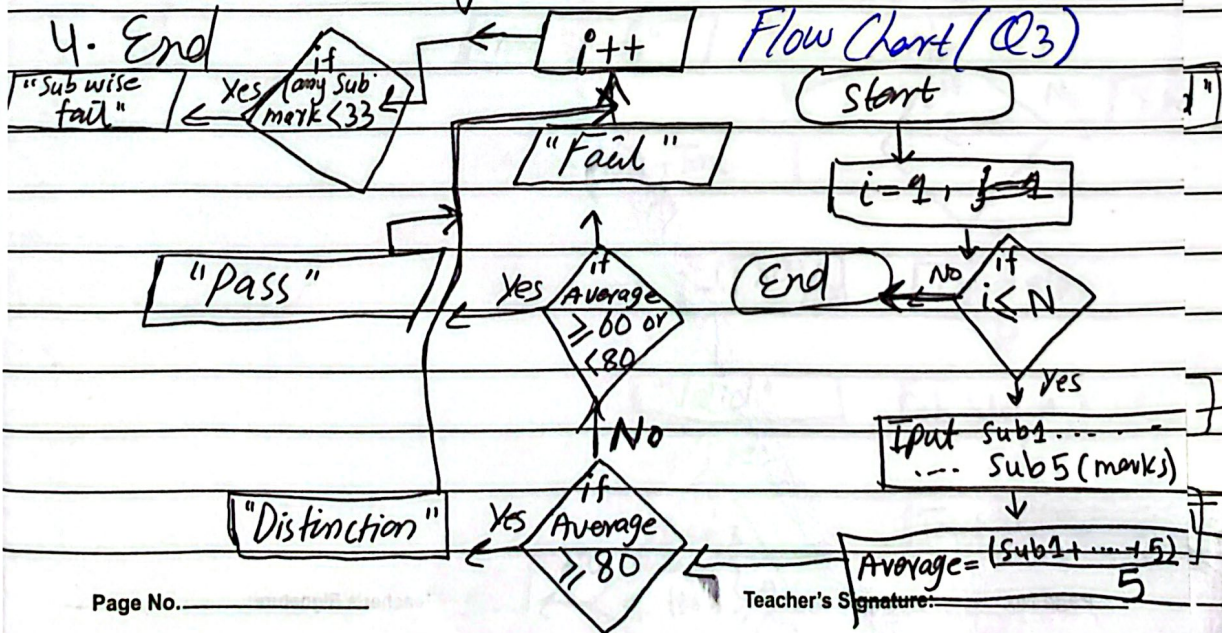
print "Fail"

~~while~~ $i++$ while $j \leq N$ if (Sub1 $\neq$ Sub2 $\neq$ Sub3 $\neq$ Sub4 $\neq$ Sub5)
are less than 33

Print "Fail - Subj Deficiency"

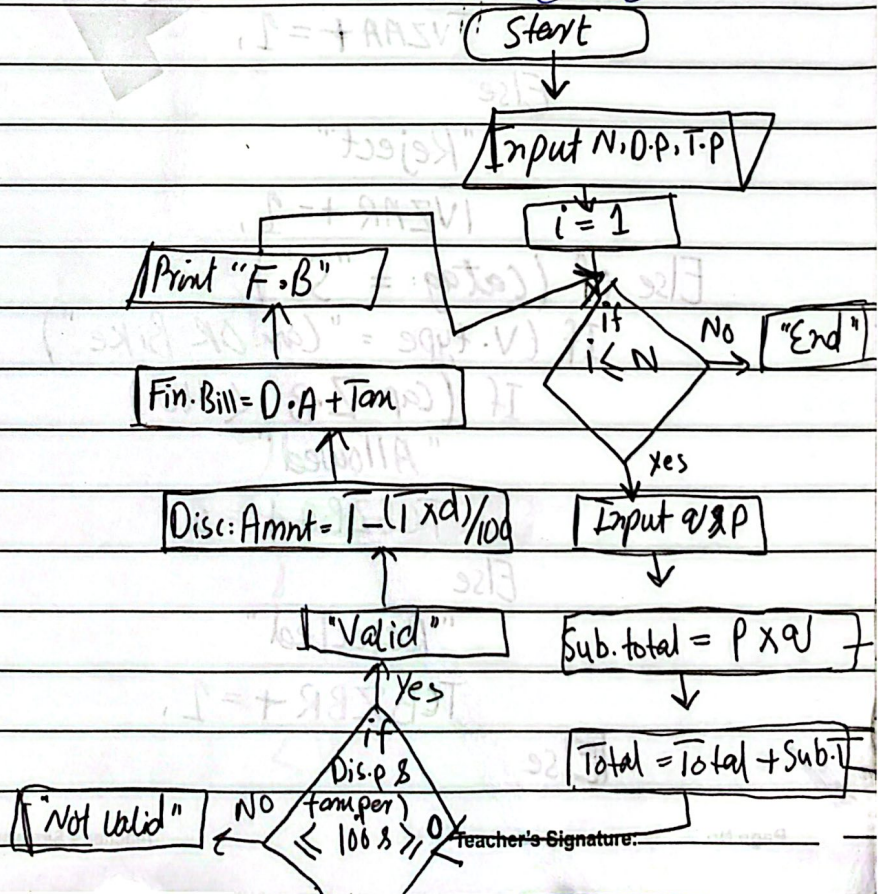
 $j++$

4. End



Question #4: (Pseudocode)

1. Begin
2. Input No. of items (N), Discounted percentage, tax percentage.
3. $i = 1$
4. while $i \leq N$
 Input quantity & price of item(1)
 Subtotal = $P \times Q$
 Total = total + Subtotal ; $i++$
5. if ((Dis: p and tax per:) ≤ 100 and ≥ 0)
 "Valid"
- Else
 "Not Valid"
6. Discounted amount = Total - (Total $\times$ d) / 100
7. Final bill = Dis: Amount + (D.A $\times$ t) / 100, t (tax percent)
8. Print "Final bill"
9. End

★ Flowchart

* TCBZAA = (Zone A Accepted)

* TCBZAR = (... Rejected)

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Question #05 (Pseudocode)

1. Begin

2. Input Valid.P, Category, Vehicle type

3. If (Valid.P = "Yes")

 If (Catag: = "F")

 If (V.type = "Car OR Bike")

 If (Cap: of Z.A < 20)

 "Allowed"

 TCBZAA += 1,

 Else

 "Reject"

 TCBZAR += 1,

 Else

 If (Cap: Z.A ≤ 18)

 "Allowed"

 TVZAA += 1,

 Else

 "Reject"

 TVZAR += 1,

 Else if (catag: = "S")

 If (V.type = "Car OR Bike")

 If (Cap: Z.B < 40)

 "Allowed"

 TCBZBA += 1,

 Else

 "Rejected"

 TCBZBR += 1,

 Else

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If (cap: Z.C ≤ 13)

"Allowed"

TVZCA += 1

Else

"Rejected"

TVZCR += 1

Else if (catag: = "V")

If (V-type = "Car OR Bike")

If (cap: Z.C < 15)

"Allowed"

TCBZCA += 1,

Else

"Reject"

TCBZCR += 1,

Else

If (cap: Z.C ≤ 13)

"Allowed"

~~Else~~ TVZCA += 1.

7

Else

"Reject"

TVZCR += 1,

Else

If (V.T = "Emergency")

~~If (cap: Z.A ≤ 18) OR ≤ 20)~~If (cap: Z.A < 20)

"Allowed"

TCBZAA += 1,

Else if (cap: Z.B < 40)

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"Allowed"

$TCBZBA += 1$

Else if (Cap: Z.C < 15)

"Allowed"

$TCBZCA += 1$

~~Else if (Cap: Z.A < 18)~~

~~"Allowed"~~

~~$TVZAA += 1$~~

Else

"All Zones are full"

"Allowed"

Else {

"Rejected, Not valid permit"

$IRNVP += 1$

4. $TAVS = TAVZAA + TVZBA + TVZCA$

5. $TRVS = TVZAR + TVZAR + TVZCR$

6. $TVPRS = \text{Print } "TVPRS"$

7. $TNCB = \text{Print } "TNCB"$

8. $TAVA = \text{Print } "TAVA"$

9. $RCZA = \text{Total Capacity (20)} - TCBVZAA$

10. $RCZB = 40 - TCBVZBA$

11. $RCZC = 15 - TCBVZ$

~~12. If~~

12. $TOZAp = TCVBZAA$

13. $TOZBp = TCVBZBA$

14. $TOZCp = TCVBZCA$

15. If ($TOZAp > TOZBp$)

"Occupancy in A is greater"

Else if ($TOZBp > TOZCp$)

"Occupancy in B is greater"

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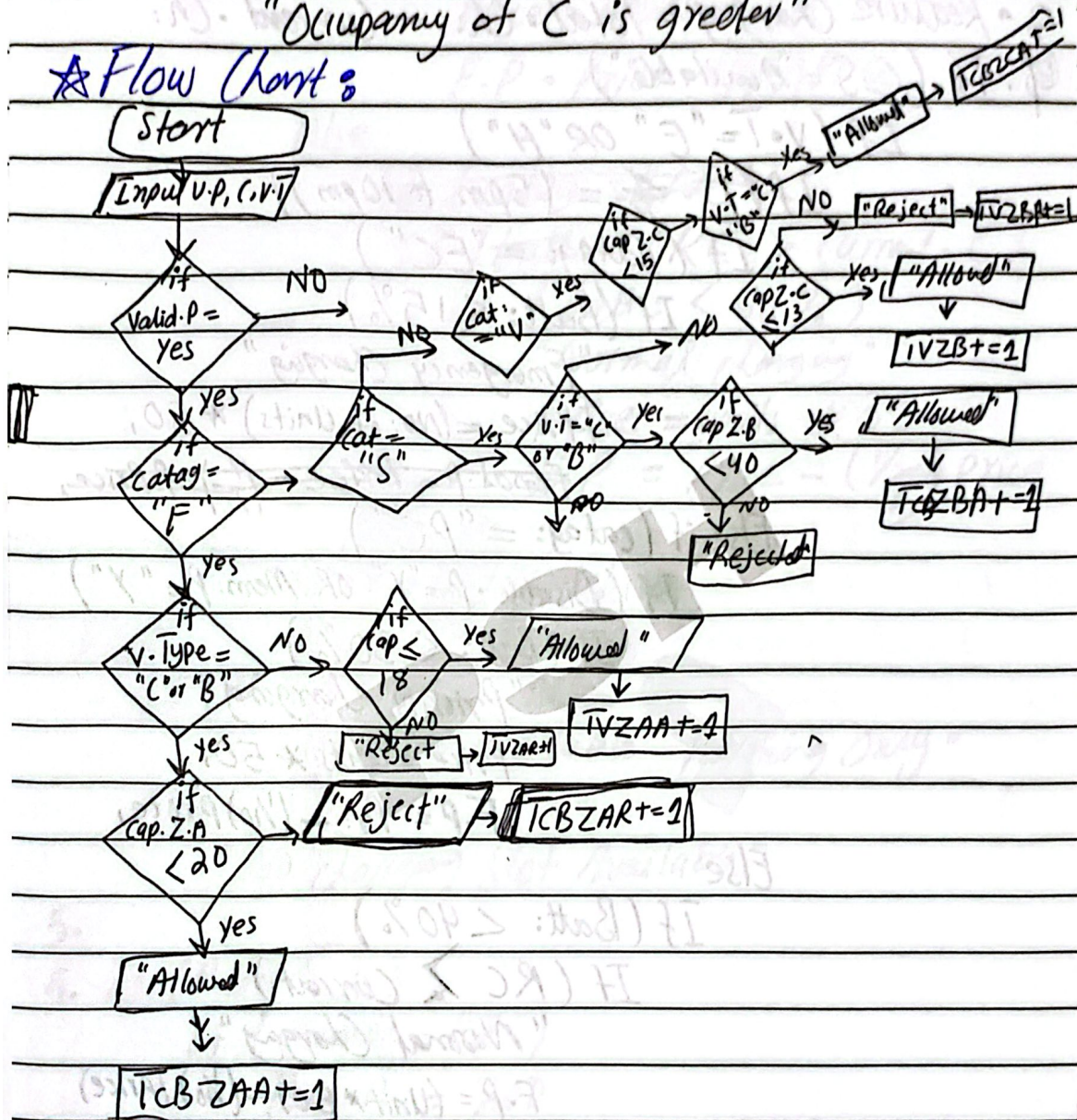
Else if ($TOZAp = TOZBp = TOZCp = 100\%$)

"Entire campus parking is full"

Else

"Occupancy of C is greater"

★ Flow Chart :



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* Question #06 (Pseudocode)

1. Begin

2. Input (V.T, C.B.L, R.B.L, P.Hrs, C.T, P.Memb.) Prio: (Y/N), CS;

3. Require Charging = Total.Ch: - Current.Ch:

If (CS = "Available")

 If (V.T = "E" OR "H")

 If ($t \neq$ (5pm to 10pm))

 If (catag: = "EC")

 If (Batt: $\leq$ 15%)

 "Emergency Charging"

 F.Price = (No: of Units) * 50,

~~Final.P = Price = (1/10) of Price,~~

 Else if (catag: = "PC")

 If (Disable.P = "Y" OR Mem.P = "Y")

 If (Batt: $\leq$ 30%)

 "Priority Charging"

 Price = units * 50,

 F.P = Pr: - (1/10) Price,

 Else

 If (Batt: $<$ 40%)

 If (RC $\geq$ Current)

 "Normal Charging"

 F.P = (Unit * 50) - (1/10) (Price)

 Else if ($t \neq$ (5pm to 10pm))

 If (catag: = "EC")

 If (Batt: $\leq$ 15%)

 "Emergency charging"

 F.P = (units * 35).

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Else if (catag: = "PC")

If (batt: $\leq$ 30%)

"Priority charging"

Price = units * 35,

F.P = Price - ($\frac{1}{5}$) Price

Else

~~If (catag: = "NE")~~

If (batt: $\rightarrow$ (RC > (current.C))

If (batt: < 40%)

"Normal charging"

Price = units * 35

F.P = Price - ($\frac{1}{5}$) Price

Else

"Not Electric Vehicle"

Else

If (V.T = "H")

"Charging unavailable - parking Only"

Else

"No charging Slot Available"

5.

5. If (P.Hrs $\leq$ 2)

If (Memb: = "X")

Park: Pr: = 200

Disc: Pr = Pr - ($\frac{1}{5}$) Price

Else

Price = 200,

Else if (Hrs > 2 and Hrs $\leq$ 5)

If (Member = "X")

Price = 320

Else

price = 400

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Else if (Hrs > 5)
If (member = "Y")
Price = 500.

Else
P.Price = 700.

6. If (Hrs > 8)

"Long-stay warning: please relocate your Veh
Else

"Standard parking duration"

7. Print ((Charging Cost) = OR (Final Price)

8. Print (Vehicle type)

9. Print (Current battery percentage)

10. Print (Required Charging)

11. Print (Parking Cost)

12. Final Payable amount = Charging(F.Price) + Parking * P